

SUGAM LAMSAL

Aerospace Engineer

+977 9840316153 · sugamlamsal@gmail.com · sugamlamsal.com.np · Kathmandu, Nepal

EDUCATION

Bachelor of Engineering — Aerospace Engineering

2021 – 2025

IOE Pulchowk Campus, Tribhuvan University, Lalitpur, Nepal

- Aggregate: 77.66% · IELTS: Overall Band 7.5
- Relevant Coursework: Fluid Mechanics, Aerodynamics, Compressible Aerodynamics, Computational Fluid Dynamics, Numerical Methods, Flight Dynamics, Control Systems, Propulsion, UAV Synthesis, Avionics

CONFERENCE PAPERS

- AIAA SciTech Forum 2027**, Orlando, FL (Accepted for presentation, Session GNC-43 — Missile and Trans-Atmospheric Vehicle GNC I)
Lamsal, S., Tajpuriya, M., Paudel, S., Lamichhane, S., Bhattarai, S. "*A Low-Cost Active Fin Control System for Sounding Rockets: Design and Flight Validation.*"
- SAROD 2024** — Symposium on Applied Aerodynamics and Design of Aerospace Vehicles, India
Lamsal, S., Paudel, S., Lamichhane, S., Tajpuriya, M. "*Optimization and Comparison of Subsonic and Hypersonic Control Surfaces of Rockets.*" Parametric missile fin geometry optimization at Mach 0.6 and Mach 5 using k- ω SST CFD (ANSYS Fluent) and Response Surface Optimization (ANSYS DesignXplorer); found a 4% L/D improvement at hypersonic conditions from an optimized sweep angle ($66.5^\circ \rightarrow 68.8^\circ$)

RESEARCH EXPERIENCE

Research Intern — Propellant Research Lab, IIT Bombay

Oct 2024 – Nov 2024

Supervisor: Prof. Nagendra Kumar, Department of Aerospace Engineering

- Ran Crawford Bomb (strand burner) tests across HTPB, TP11 solid propellant (with and without PTHF binder), gas-generating propellant, wax, and slurry systems at chamber pressures from 1–70 bar; fitted results to the empirical burn rate law $r = aP^n$ to extract burn rate coefficient and pressure index; ran temperature sensitivity tests at 70°C
- Conducted Constant Volume (CV) test ballistic characterization — pressure-time and temperature-time analysis of gun and gas-generating propellants — including igniter mass optimization and residual mass tracking
- Performed Quench Bomb rapid-depressurization studies of binder-melting characteristics and combustion quenching behavior on TP11 and slurry propellants
- Prepared and cast HTPB fuel grains for hybrid rocket motor use, including vacuum-degassing and oven-curing cycles to eliminate air-bubble defects; assisted in hybrid rocket motor firing tests and a scaled L-70 artillery firing test

OpenFOAM CFD Research Intern — FOSSEE, IIT Bombay

Feb 2025 – Jun 2025

Project: Exploring Immersed Boundary Method in OpenFOAM | Unsteady Airfoil Aerodynamics

- Implemented the Immersed Boundary Surface discrete-forcing method in foam-extend 4.1, adding a forcing term to the momentum equations on a fixed background Cartesian mesh to enforce no-slip at a moving body surface.
- Simulated sinusoidally plunging elliptical airfoil at $Re = 500$ at reduced frequency $k = 6.283$ across multiple plunging amplitudes
- Validated the discrete-forcing IB formulation against published continuous-forcing results as a cross-method check, and validated lift and thrust coefficients against literature
- Studied leading-edge vortex formation, shedding, and convection on the plunging airfoil
- Ran mesh sensitivity and timestep independence studies to verify numerical convergence; post-processed results using Paraview and MATLAB on a Linux workflow

Research Submission — Rotorcraft Aerodynamics and Aeroacoustics Lab, Oklahoma State University 2026 – Present

Supervisor: Dr. Sicheng Kevin Li

- Implemented the full four-subsystem Beddoes-Leishman dynamic stall model in Fortran 90, covering attached-flow indicial response, trailing-edge separation, leading-edge vortex formation and convection, and pitching moment and drag
- Validated CN, CM, CD hysteresis loops against Leishman and Beddoes (1989) light- and strong-stall experimental data; implemented a smooth vortex-onset factor replacing the model's original hard piecewise switch
- Corrected multiple discrepancies in the lab's trailing-edge noise code (Amiet/Roger-Moreau formulation) through direct equation-to-code validation against a published Journal of the American Helicopter Society formulation
- Currently collaborating on the acoustic scaling of small and large rotors, validating model output against Brooks experimental trailing-edge noise data using Buckingham Pi non-dimensional scaling analysis

ROCKET & AIRCRAFT DESIGN PROJECTS

Integrated Control System for Active Fin-Controlled Rocket Stabilization *Final Year Project, 2024–2025*

- Designed and implemented a PID-based active fin control system; controller processed Kalman-filtered IMU data (BNO085 + MPU6050) to generate real-time servo deflection commands across pitch and roll axes
- Developed a physics-based 6-DOF MATLAB/Simulink model incorporating actuator lag, sensor noise, discrete sampling, rate limiting, and variable-mass dynamics for PID gain tuning; generated aerodynamic coefficient tables (CL, CD, Cm vs. AoA) via XFLR5 (VLM) and ANSYS Fluent (k- ϵ , 1.4M elements)
- Designed and fabricated a custom flight computer PCB (KiCad) built around a Teensy 4.1 (primary) and ESP32 (backup), with BMP180 barometer, GPS, and LoRa telemetry for real-time downlink
- Validated pitch control authority ($< 2^\circ$ steady-state oscillation) and roll disturbance rejection ($> 300^\circ/\text{s}$) through wind tunnel testing; confirmed active roll stabilization within $\sim 5^\circ$ error margin in flight test; post-flight sensor data compared against 6-DOF Simulink predictions

SPLOOSH — Amphibious Aircraft Design

2024

18-passenger amphibious aircraft, twin-engine, high-wing configuration

- Led sizing and matching-plot analysis to establish the design point, and full dynamic stability evaluation across all four classical modes (phugoid, short period, Dutch roll, spiral), reporting frequency and subsidence ratio for each
- Determined the neutral point from elevator-trim-vs-CL data across multiple CG positions and built a CG travel diagram to confirm static margin throughout the mission from empty weight to MTOW
- Ran a bank-angle handling quality test against a 30° target and rated the aircraft on the Cooper-Harper scale (rating of 3, mild deficiencies with minimal pilot compensation required)
- Computed flight and gust envelopes, stall/climb/descent characteristics, and separate takeoff and landing distance analysis for ground and water operations, using XFLR5, OpenVSP, and X-Plane simulation data post-processed in MATLAB

Project Trishul — High-Power Sounding Rocket

2023–2024

Two rockets launched; apogees of $\sim 2,600$ m and $\sim 1,800$ m achieved

- Led avionics integration: custom KiCad PCB with Teensy 4.1, MPU6050 IMU, BMP180, GPS, and LoRa telemetry for real-time flight data acquisition; redundant MOSFET-controlled black powder ejection for dual-parachute recovery
- Conducted aerodynamic design in OpenRocket, fin flutter analysis by Martin's method, and static motor testing (KNDX propellant, three-grain motor) validating impulse, thrust, and burn time against OpenMotor predictions
- Fabricated fiberglass composite rocket body with vacuum-bagged, tip-to-tip reinforced fin construction; compared post-flight telemetry against OpenRocket-simulated trajectory

AIAA Design/Build/Fly Competition

2024

Rank: 21st (Proposal), 38th (Overall)

- Aircraft conceptual and preliminary design for UAM missions (delivery, medical transport, urban taxi); aerodynamic sizing via XFLR5, static and dynamic stability analysis, drag estimation via ANSYS Fluent
- Structural FEM analysis of wing and landing gear in SolidWorks; payload-endurance optimization for each mission type; carried design through manufacturing and flight testing

CFD Analysis of Winglet Cant Angle Optimization

2023

- Co-authored a study on winglet cant angle optimization for aircraft aerodynamic performance; designed wing and winglet geometry (NACA 2412, AR 7, 1.4 m span) in CATIA V5 at four cant angles (0° , 45° , 60° , 90°)
- Built a 3D unstructured tetrahedral mesh in ANSYS Workbench with 10 boundary-layer inflation layers and local refinement around the wing/winglet; solved transonic flow ($M = 0.8395$) in ANSYS Fluent using the Spalart-Allmaras turbulence model across $\alpha = 0\text{--}9^\circ$
- Computed CL, CD, and L/D across the angle-of-attack sweep for each cant angle; identified 45° as the optimal configuration for maximum lift-to-drag ratio

INDUSTRY EXPERIENCE

In-Service Engineering Intern — Airbus India

Jul 2025 – Jan 2026

- Performed fatigue life analysis and repair design for the LR main landing gear (MLG) support rib and assisted on fatigue justification for repairs to LR MLG support rib lugs; produced a repair sketch for a trailing edge overhang panel
- Built repair-engineering toolkits supporting automated Substantiation/Structural Approval Sheet (SAS) generation, a Generic Repair Justification database, and a Check-stress database
- Developed a Python-based automated wing-skin panel bolt location diagram tool with auto-annotation, integrating Python, MS Excel, and HTML for engineering documentation automation
- Used Matplotlib for dataset visualization and post-processing across engineering projects; built a Google Site to centralize daily trackers, planners, and toolkits for the team

TECHNICAL SKILLS

- **CFD & Simulation:** OpenFOAM (foam-extend 4.1) · Immersed boundary methods · ANSYS Fluent ($k-\omega$ SST, $k-\epsilon$, Spalart-Allmaras) · ANSYS DesignXplorer (RSO) · XFLR5 · OpenVSP · OpenRocket / OpenMotor · X-Plane
- **Propellant Testing & Internal Ballistics:** Constant Volume test, Crawford Bomb (strand burner) test, Quench Bomb test, burn rate law fitting ($r = aP^n$), hybrid motor grain casting
- **Flight Dynamics & Control:** PID control · Kalman filtering · 6-DOF simulation (MATLAB/Simulink) · Indicial/semi-empirical aerodynamic modeling
- **Embedded & Avionics:** Teensy 4.1 · KiCad PCB design · IMU / barometer integration · LoRa telemetry · MOSFET ejection circuits
- **Programming & Computing:** Fortran 90 · Python · Embedded C · MATLAB/Simulink · Linux
- **CAD & Design:** CATIA V5 · SolidWorks · Fusion 360

PRESENTATIONS & ACTIVITIES

- SpaceCon 2025 — Presented "Integrated Control System for Active Fin-Controlled Rocket Stabilization and Guidance"
- Boeing AeroModelling Challenge, TechFest 2023, IIT Bombay — Designed, manufactured, and flew a fixed-wing RC aircraft with accurate mid-air payload drops
- Organized OpenRocket and XFLR5 workshops for undergraduate students at Pulchowk Campus